Fourier series

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Basic formula :

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\int uv\,dx = u\int v\,dx - \int\left\{\frac{d}{dx}(u)\int v\,dx\right\}dx$$

Advance :

$D = \frac{d}{dx}$ $\quad$ V_1, V_2, V_3

$$\int uv\,dx = uV_1 - u'V_2 + u''V_3 - \cdots$$

$$\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}\left[a\sin bx - b\cos bx\right]$$

$$\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}\left[a\sin bx + b\cos bx\right]$$

- $\sin n\pi = 0$
- $\cos n\pi = (-1)^n$
- $\cos 2n\pi = 1$
- $\sin 2n\pi = 0$

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Fourier Serise Formula :

$$f(x) = a_0 + \sum_{n=1}^{\alpha} [a_n \cos nx + b_n \sin nx]$$

$$\alpha < x < \alpha + 2\pi$$

Then,

$$a_0 = \frac{1}{2\pi} \int_{\alpha}^{\alpha+2\pi} f(x)\, dx$$

$$a_n = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} f(x) \cos nx\, dx$$

$$b_n = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} f(x) \sin nx\, dx$$

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$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right]$$

for, $-l < x < l$

$$a_0 = \frac{1}{2l} \int_{-l}^{l} f(x)\, dx$$

$$a_n = \frac{1}{l} \int_{-l}^{l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$b_n = \frac{1}{l} \int_{-l}^{l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

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Practic Question:

Find the fourier series of the $f(x) = e^{ax}$ in $(0, 2\pi)$

We know

$$f(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$$

$$e^{ax} = a_0 + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$$

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(x)\, dx = \frac{1}{2\pi} \int_0^{2\pi} e^{ax}\, dx$$

$$= \frac{1}{2\pi} \left[\frac{e^{ax}}{a} \right]_0^{2\pi}$$

$$= \frac{1}{2\pi} \left[\frac{e^{2a\pi}}{a} - \frac{1}{a} \right]$$

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$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\,dx$$

$$= \frac{1}{\pi}\int_0^{2\pi} e^{ax}\cos nx\,dx$$

$$= \frac{1}{\pi}\left[\frac{e^{ax}}{a^2+n^2}(a\cos nx + n\sin nx)\right]_0^{2\pi}$$

$$= \frac{1}{\pi}\left[\frac{a e^{ax}}{a^2+n^2}\cos nx + \frac{n e^{ax}}{a^2+n^2}\sin nx\right]_0^{2\pi}$$

$$= \frac{1}{\pi(a^2+n^2)}\left[e^{ax}\cos nx + e^{ax}\sin nx\right]_0^{2\pi}$$

$$= \frac{1}{\pi(a^2+n^2)}\left[e^{2\pi a} - 1 - a\cdot 1\right]$$

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$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\,dx$$

$$= \frac{1}{\pi}\int_0^{2\pi} e^{ax}\sin nx\,dx$$

$$= \frac{1}{\pi}\left[\frac{e^{ax}}{a^2+n^2}(a\cos nx - n\sin nx)\right]_0^{2\pi}$$

$$= \frac{1}{\pi}\left[\frac{ae^{ax}}{a^2+n^2}\cos nx - \frac{ne^{ax}}{a^2+n^2}\sin nx\right]_0^{2\pi}$$

$$= \frac{1}{\pi(a^2+n^2)}\left[ae^{ax}\cos nx - ne^{ax}\sin nx\right]_0^{2\pi}$$

$$= \frac{1}{\pi(a^2+n^2)}\left[e^{2\pi a}\cdot 0 - ne^{2\pi a}\cdot 1\right] - \left[n\cdot 1\right]$$

$$= \frac{-b_n}{\pi(a^2+n^2)}\left[e^{2\pi a}-1\right]$$

Put the value in the main

$$e^{ax} = \frac{1}{2\pi}\left[\frac{e^{2\pi a}-1}{a}\right] + \sum_{n=1}^{2}\left[\frac{a}{\pi(a^2+n^2)}(e^{2\pi a}-1)\cos nx + \frac{n}{\pi(a^2+n^2)}(1-e^{2\pi a})\sin nx\right]$$

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Formula :

$$f(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$$

$-\pi < x < \pi$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx\, dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx\, dx$$

even function (+)

$F(x) = f(-x)$

x^2, $\cos x$, $x \sin x$ (− − = +)

$$a_0 = \frac{2}{2\pi} \int_{0}^{\pi} f(x)\, dx$$

$$a_0 = \frac{1}{\pi} \int_{0}^{\pi} f(x)\, dx$$

$$a_n = \frac{2}{\pi} \int_{0}^{\pi} f(x) \cos nx$$

$b_n = 0$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

odd function (−)

$F(x) = -f(-x)$

x, $\sin x$ $x \cos x$ (+ − = −)

$a_0 = 0$

$a_n = 0$

$$b_n = \frac{2}{\pi} \int_{0}^{\pi} f(x) \sin nx\, dx$$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

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(ii) find the fourier serise of the $f(x)=x^2, \quad -\pi<x<\pi$

As an even function we know

$$f(x)=a_0+\sum_{n=1}^{\infty} a_n \cos nx$$

$$x^2=a_0+\sum_{n=1}^{\infty} a_n \cos nx$$

then $b_n=0$

$$a_0=\frac{1}{\pi}\int_0^{\pi} f(x)\,dx=\frac{1}{\pi}\int_0^{\pi} x^2\,dx$$

$$=\frac{1}{\pi}\left[\frac{x^3}{3}\right]_0^{\pi}$$

$$=\frac{1}{\pi}\times\frac{\pi^3}{3}$$

$$=\frac{\pi^2}{3}$$

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$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx$$

$$= \frac{2}{\pi}\int_0^{\pi} x^2 \cos nx\,dx$$

$$= \frac{2}{\pi}\left[x^2\left(\frac{\sin nx}{n}\right) - 2x\left(\frac{-\cos nx}{n^2}\right) + 2\left(\frac{-\sin nx}{n^3}\right)\right]$$

$$= \frac{2}{\pi}\left[2\pi\left\{\frac{(-1)^n}{n^2}\right\}\right] = \frac{4(-1)^n}{n^2}$$

Put the value in the formula

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

$$x^2 = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2}\cos nx \quad (1)$$

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Find the fouries ~~series~~ series of the following function

(i) $f(x) = x \quad -\pi < x < \pi$

It is an odd function there

$a_0 = 0, \quad a_n = 0$

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x) \sin nx \, dx$$

$$= \frac{2}{\pi}\int_0^{\pi} x \sin(nx)\, dx$$

$$= \frac{2}{\pi}\left[-\frac{x\cos(nx)}{n} + \frac{\sin(nx)}{n^2}\right]_0^{\pi}$$

$$b_n = \frac{2(-1)^{n+1}}{n}$$

Put the value

$$f(x) = \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n} \sin(nx)$$

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$$x = 2\left[\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \frac{\sin 4x}{4} + \cdots\right]$$

(1)

finite fourier sine and cosine

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finite fourier sine ~~and cosine~~ transform

The finite fourier sine transform of $f(x)$ is $0<x<c$ is defined by

$$f_s(n) = \int_0^c f(x) \cdot \sin \frac{n\pi x}{c} \, dx \quad [n = \text{integer}]$$

Also, the function $f(x)$ [inverse finite fourier sine transform of $f_s(n)$] is given by

$$f(c) = \frac{2}{c} \sum_{n=1}^{\infty} f_s(n) \cdot \sin \frac{n\pi x}{c}$$

Finite fourier cosine transform:

The finite fourier cosine transform of $f(x)$ is $0<x<c$ is given by

$$f_c(n) = \int_0^c f(x) \cos \frac{n\pi x}{c} \, dx$$

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Also, the function $f(x)$ [inverse finite fourier cosine transform of $f_c(n)$] is given by

$$F(x) = \frac{1}{c} f_c(0) + \frac{2}{c} \sum_{n=1}^{\infty} f_c(n) \cos \frac{n\pi x}{c}$$

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Find the finite fourier series sine and cosine transformation of the function

(i) $f(x) = 2x \quad 0 < x < 4$

(ii) $f(x) = x^2 \quad 0 < x < 3$

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Fourier Integral

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Fourier integral of an even function

$f(x) = f(-x)$ [cosine]

$$f(x) = \frac{2}{\pi} \int_0^{\alpha} \cos ux \left[\int_{-\alpha}^{\alpha} f(t) \cos ut \, dt \right] du$$

Fourier integral of an odd function

$f(-x) = -f(x)$ [sine]

$$f(x) = \frac{2}{\pi} \int_0^{\alpha} \sin ux \left[\int_0^{\alpha} f(t) \sin ut \, dt \right] du$$

$$= \frac{2}{\pi} \int_0^{\alpha} \left[\int_0^{\alpha} f(t) \sin ut \, dt \right] \sin ux \, dx$$

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Fourier Integral Theorem

$$f(x) = \frac{1}{\pi} \int_0^{\infty} [A(u)\cos(ux) + B(u)\sin(ux)]\, du$$

where

cosine coefficient:

$$A(u) = \int_{-\infty}^{\infty} f(t)\cos(ut)\, dt$$

sine coefficient:

$$B(u) = \int_{-\infty}^{\infty} f(t)\sin(ut)\, dt$$

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Find the fourier sine integral of

$$f(x) = \begin{cases} x, & 0<x<a \\ 0, & x>a \end{cases}$$

fourier sine integral formula

$$f(x) = \frac{2}{\pi}\int_0^{\alpha} \sin ux \left[\int_0^{\alpha} f(t)\sin ut\, dt\right] du$$

$f(t)=0$ when $t>a$. Therefore, we only need to integrate from 0 to a. the limit change 0 to a

We get

$$f(x) = \frac{2}{\pi}\int_0^{\alpha} \left[\int_0^{a} x \sin ut\, dt\right] \sin ux\, du$$

Now slove the

$$I = \int_0^{a} x \sin ut\, dt$$

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$$I = \left[-\frac{x\cos(ux)}{u} + \frac{\sin(ux)}{u^2} \right]_0^a$$

$$I = \frac{\sin(ua)}{u^2} - \frac{a\cos(ua)}{u}$$

Put the value in the formula

$$f(x) = \frac{2}{\pi} \int_0^\infty \left[\frac{\sin(ua)}{u^2} - \frac{a\cos(ua)}{u} \right] \sin(ux)\, dx \quad (1)$$

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using the fourier cosine integral show that $x > 0$

$$\frac{\pi}{6} e^{-3x} = \int_0^\infty \frac{\cos ux}{u^2+9} \, dx$$

the cosine formula:

$$f(x) = \frac{2}{\pi} \int_0^\infty \left[\int_0^\infty f(x) \cos ut \, dt \right] \cos ux \, dx$$

there limit upper $= \infty$, lower $= 0$

Now,

$$I = \int_0^\infty f(x) \cos(ut) \, dt$$

$$I = \int_0^\infty e^{-3t} \cos(ut) \, dt$$

$$I = \frac{3}{u^2+9}$$ [first Do integration then put limit]

thus eqn the given value

$$f(x) = \frac{2}{\pi}\int_0^{\infty}\left[\frac{3}{u^2+9}\cos(ux)\,du\right]$$

$$\Rightarrow e^{-3x} = \frac{6}{\pi}\int_0^{\infty}\frac{3}{u^2+9}\cos ux\,du$$

$$\frac{\pi}{6}e^{-3x} = \int_0^{\infty}\frac{3}{u^2+9}\cos(ux)\,du$$

(proved)